Homework 1

1. Let $f(x) = 1$ if x is rational, $f(x) = 0$ if x is irrational. Show that f is not integrable on any interval.

2. Let x_k be a convergent sequence in $\mathbb{R}$. Write down the definition of content zero and show that the set $\{x_1, x_2, ...\}$ has zero content.

3. Let f be an integrable function on $[a, b]$. Suppose that $f(x) \geq 0$ for all x and there exists $c \in [a, b]$ such that f is continuous at c and $f(c) > 0$. Show that $\int_a^b f(x)dx > 0$.

4. If $f : [a, b] \to \mathbb{R}$ is bounded on $[a, b]$ and is continuous at all except finitely many points in $[a, b]$. Show that f is Riemann integrable on $[a, b]$.

5. Let $f : [a, b] \to \mathbb{R}$ be Riemann integrable on $[a, b]$. Show that $|f|$ is Riemann integrable and $|\int_a^b f(x)dx| \leq \int_a^b |f(x)|dx$.